



HireHub – AI Resume-Matching Job Portal

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Abstract

- Job portal where candidates' resumes are AI-parsed into skills and matched to recruiters' jobs; built with React, Express and MongoDB.
- Faster, skill-based hiring for both sides.

Problem Statement & Objectives

- Problem: Portals rely on keyword search and manual resume screening, not skill fit.
- Objective 1: Auto-parse resumes into skills
- Objective 2: Score every job by skill match
- Objective 3: Track hiring from apply to offer
- Scope: Candidate, recruiter, admin; web app

Existing vs Proposed System

Feature	Existing System	Proposed System
Resume	Manual form filling	AI + OCR extraction
Matching	Keyword search only	Skill match score %
Tracking	Status often hidden	Live 7-stage status
Interviews	Arranged by email	Schedule + feedback
Limitation / Gap	Recruiters screen each resume by hand	Match score saved on every application

Feasibility Study

- Technical: One JavaScript stack (React, Node.js, MongoDB); all tools freely available.
- Economic: Open-source stack; AI parsing is optional, with a free rule-based fallback.
- Operational: Role-based dashboards and one-click resume upload; easy to maintain.
- Schedule: Built module by module within the semester timeline.

Results & Testing

- 31 REST routes over 7 MongoDB collections
- CRUD with JWT role guards and 400/409 checks

Conclusion

- Working AI-assisted portal for all 3 roles
- Parsing, scoring and tracking all delivered

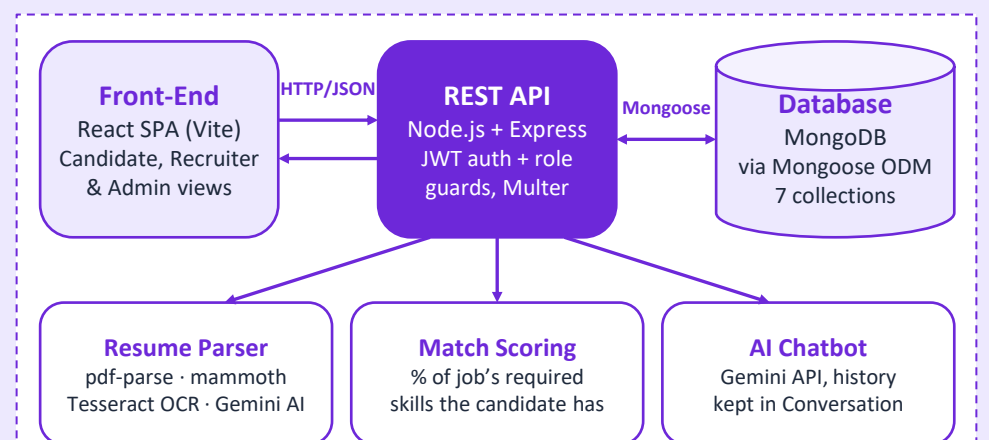
Future Scope

- Semantic (embedding) job matching
- Chatbot grounded in live user data

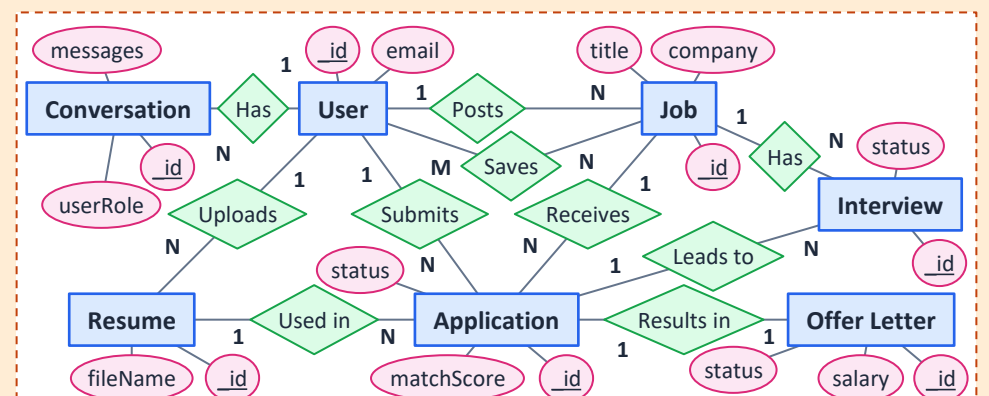
Tools & Technologies

- Front-End: React 19 + Vite, React Router
- Back-End: Node.js + Express + REST APIs; Gemini API, pdf-parse, Tesseract OCR
- Database: MongoDB with Mongoose ODM
- Tools: Git & GitHub, VS Code, npm, nodemon

System Architecture



ER Diagram & Database Design



Implementation Screenshots

